

The empirical recurrence for OEIS A237476: recovering a conjecture that is not written down

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7 September 2026

Abstract

OEIS A237476 records “Empirical recurrence of order 73”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 178$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 73, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 73 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A237476 is “Number of $(n+1) \times (3+1)$ 0..2 arrays with the maximum plus the lower median of every 2×2 subblock equal.”. It has offset 1 and begins

1337, 19175, 273756, 4384979, 71061267, 1184249828, 19835148309, 334897261287, ...

The entry states:

Empirical recurrence of order 73 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 10:12:00, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 178$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 178$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 356$ exact terms of the model returns order 73 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{73} c_i a(n-i),$$

c_1	24	c_{20}	132937549899031809	c_{39}	189265217629044248029334	c_{58}	5669544
c_2	216	c_{21}	−57223965741744522	c_{40}	68429978480049972687950	c_{59}	1973425
c_3	−8276	c_{22}	−1324468730941933534	c_{41}	−328946443664790007682551	c_{60}	−1298509
c_4	−1861	c_{23}	1068475609904223217	c_{42}	−60142235348902158951529	c_{61}	−228864
c_5	1166736	c_{24}	10539801712410825456	c_{43}	467597635368710546414862	c_{62}	2090074
c_6	−3191279	c_{25}	−11762454331311616768	c_{44}	1853633833016940955471	c_{63}	977130
c_7	−91235323	c_{26}	−67512288405742533507	c_{45}	−541795401357281641764800	c_{64}	−222520
c_8	402333178	c_{27}	93552613049106259400	c_{46}	41609493021211604296382	c_{65}	120860
c_9	4480606213	c_{28}	349841627389919200693	c_{47}	508925504515936762736291	c_{66}	140306
c_{10}	−25766386496	c_{29}	−572731685896476840451	c_{48}	−88067228082807638562910	c_{67}	−19735
c_{11}	−147151963954	c_{30}	−1470149915261604564869	c_{49}	−384524615318262807103153	c_{68}	−4018
c_{12}	1058614532664	c_{31}	2778362090494465163199	c_{50}	97809824091929438892961	c_{69}	10066
c_{13}	3317970084353	c_{32}	5008537685104250658538	c_{51}	231151296852062265828100	c_{70}	−396
c_{14}	−30543061732853	c_{33}	−10849881033010974200915	c_{52}	−75236200373596837632797	c_{71}	−151
c_{15}	−50598457068202	c_{34}	−1378013124588880250708	c_{53}	−108878797974707354158960	c_{72}	1374
c_{16}	650242307521613	c_{35}	34423586908151572264928	c_{54}	42614253121620477412880	c_{73}	−30
c_{17}	461230172021974	c_{36}	30341252295210587029509	c_{55}	39314843400498432782476		
c_{18}	−10540791350189129	c_{37}	−89201129692217400669204	c_{56}	−18032354966914111409296		
c_{19}	−558909097815716	c_{38}	−52472198117878266766311	c_{57}	−10525600761768982851384		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 73, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $73 + 4$ terms removed returns order 73 as well, so $\deg q_{\min} = 73 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $73 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 73 that this sequence satisfies — and that family turns out to have exactly one member.

References

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